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Low-carbon urban transportation: Optimizing mechanical systems for sustainable electric bus and BRT deployment in Kampala

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ABSTRACT

Kampala faces increasing congestion, air pollution, and dependence on fossil fuels, driven by widespread reliance on diesel minibuses and motorcycle taxis. Existing models—KAMPALA-TIMES, KLAP-TIMES, and GKMA-TIMES–CGE—show strong potential for electrified mass transit to reduce emissions, change commuter behavior, and boost macroeconomic welfare. However, these studies assume electric-bus reliability without examining the mechanical conditions needed to achieve their projected outcomes. This study combines system-level modeling insights with vehicle-level engineering analysis to identify key mechanical factors necessary for the successful deployment of electric Bus Rapid Transit (e-BRT) in Kampala. It considers drivetrain torque for steep gradients, battery thermal management in hot equatorial climates, and regenerative braking efficiency in traffic congestion, alongside policy, infrastructure, and grid readiness. Mechanical performance links modeling to implementation—adequate torque, thermal stability, and regenerative braking efficiency directly affect service reliability, headway adherence, fleet uptime, and lifecycle costs. These operational factors influence commuter mode choices, the realism of bottom-up pathways, and the broader economic benefits predicted in top-down scenarios. Engineering reliability must be a core policy consideration, guiding procurement standards, charging infrastructure design, and multisector coordination among KCCA, MoWT, MEMD, and Uganda's power utilities. Incorporating mechanical parameters into future bottom-up or hybrid models, combined with digital-twin testing and degradation-aware analytics, will enable Kampala to serve as a living laboratory for low-carbon mobility transitions across Sub-Saharan Africa.

Significance Statement

Decarbonizing urban transport in African cities requires more than visions or financing; it depends on mechanical reliability. This study emphasizes optimized drivetrains, hot-climate battery cooling, and regenerative braking as critical enablers of electric BRT. Building on KAMPALA-TIMES (system feasibility), KLAP-TIMES (commuter behavior), and the GKMA-TIMES–CGE hybrid (macroeconomic trade-offs), we demonstrate that adoption hinges on engineering performance linked to social and economic outcomes. By aligning optimization with policy, Kampala emerges as a living laboratory whose lessons extend to equitable, low-carbon mobility transitions across the Global South.

1. Introduction: Kampala's urban transport dilemma

Urban transport in Sub-Saharan Africa is both a driver of economic development and a source of profound environmental stress [2,21]. Kampala, Uganda's capital and economic hub, accounts for >60 % of the national GDP but suffers from chronic congestion [27], inefficient paratransit systems [24], and a rapidly rising consumption of fossil fuels [25].The daily trips in the city exceed 1.2 million, with minibus taxis (matatus) and motorcycle taxis (boda-bodas) accounting for the majority of trips [25]. While these modes provide flexible and affordable mobility, they contribute disproportionately to greenhouse gas (GHG) emissions, localized air pollution, and declining urban air quality. Without decisive intervention, transport-related emissions are projected to double by 2100 under a business-as-usual scenario globally [53], while congestion already costs the Ugandan economy an estimated 5 % of GDP annually [38].

The reliance on imported fossil fuels exacerbates balance-of-payment challenges [41], and public health burdens from $PM_{2.5}$ and NOx exposure are rising [47], disproportionately affecting low-income groups who depend on informal roadside hubs. These structural and

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socio-economic risks underscore the urgency of situating Kampala's transport dilemma within broader energy–transport modeling efforts, where system-level feasibility (KAMPALA-TIMES) and commuter behavior dynamics (KLAP-TIMES) together reveal both the scale of the challenge and the critical conditions for a viable low-carbon transition [24,25].

A system-level modeling exercise using the KAMPALA-TIMES model showed that Kampala could achieve significant emission reductions by integrating electrified metro corridors into its long-term development plans. Four scenarios—Business as Usual (BAU), Renewable Energy Contribution (REC), Renewable Expansion Pathway (REP), and Carbon Reduction Target (CRT)—demonstrated that decarbonization gains of up to 40 % are technically possible if high-capacity electric transit is implemented and adequately supported by Uganda's electricity mix. However, this metro-focused analysis highlighted system-wide trajectories and grid impacts, leaving open questions about the operational feasibility of more flexible modes, such as electric bus rapid transit (e-BRT) [25].

Complementing this systems view, the KLAP-TIMES model incorporated commuter behavior into Kampala's transportation optimization framework. By explicitly representing Value of Travel Time (VTT) and Time-Travel Financial (TTF) trade-offs, it showed that mode choice is highly sensitive to service reliability, punctuality, and affordability. The Kampala Sustainable Scenario (KSS) achieved a 50 % emissions-reduction trajectory by 2060, mainly through modal shifts toward high-capacity, electrified mass rapid transit, as commuters experienced tangible time savings and greater reliability. These findings emphasize that commuter confidence in service quality is as crucial as system design in driving low-carbon transitions [24].

Also, the GKMA-TIMES–CGE hybrid framework extended these insights by linking energy system optimization with macroeconomic feedbacks for the Greater Kampala Metropolitan Area (GKMA). The study showed that deep decarbonization scenarios, such as the Lutta pathway, not only cut the emissions intensity of GDP by over 60 % but also generated positive impacts on household welfare and GDP growth, while reallocating labor and capital across sectors [18]. This integrated lens underscores that sustainable transport planning for Kampala must simultaneously address energy supply, commuter behavior, and macroeconomic trade-offs

However, neither of these studies explicitly addressed the mechanical engineering requirements, such as drivetrain adaptation, hot-climate battery cooling, and regenerative braking optimization, that are necessary for turning Kampala's system-level visions and behavioral potentials into reliable operations. Against this backdrop, the present study fills a critical gap by examining these mechanical engineering requirements that can translate system visions and behavioral potentials into reliable e-BRT operations.

1.1. Motivation and gap

Kampala's transition to electric Bus Rapid Transit (e-BRT) presents a strategic opportunity to reduce congestion, improve air quality, and enhance mobility equity simultaneously [52]. Unlike metro systems—whose high capital costs and long construction horizons limit near-term feasibility—e-BRT infrastructure can be incrementally deployed along existing radial corridors such as Jinja, Bombo, Masaka, and Entebbe Roads. International experience from Bogotá [34], Rio de Janeiro [9], and Surabaya [55] demonstrates that well-designed BRT systems can reduce emissions by 20–40 % while improving access for low-income commuters. For Kampala, the integration of dedicated lanes, signal priority, and electric buses offers immediate gains in travel time, commuter confidence, and public health [24].

However, these gains depend fundamentally on mechanical and infrastructural readiness. Kampala's steep gradients, hot equatorial climate, and evolving grid conditions require electric buses to deliver high torque, maintain thermal stability, and recover braking energy efficiently in congested traffic. Existing studies—KAMPALA-TIMES (40 % emissions reduction potential), KLAP-TIMES (reliability-driven mode shift), and GKMA-TIMES–CGE (over 60 % GDP-intensity reduction by 2050)—highlight system, behavioural, and macroeconomic potential but do not examine the engineering conditions necessary for reliable operation. This leaves a critical gap between modeled feasibility and operational reality.

This study fills that gap by arguing that drivetrain adaptation, battery thermal management, and regenerative-braking optimization form the essential mechanical bridge linking system-level ambitions, commuter adoption, and economy-wide sustainability.

Guiding this inquiry are four research questions:

(i) How can drivetrain, cooling, and braking innovations enhance electric-bus reliability across Kampala's terrain and climate? (ii) How can these mechanical improvements, when combined with e-BRT infrastructure, connect insights from system-level (KAMPALA-TIMES) and behavior-level (KLAP-TIMES) modeling? (iii) What co-benefits in health, equity, and energy security emerge when e-BRT systems gain operational credibility? (iv) How can Kampala be positioned as a "living laboratory," integrating bottom-up/ hybrid synergies, digital twins, and engineering principles to advance mobility transitions across Africa and the broader Global South?

These research questions highlight the need for an integrated conceptual framework that links system feasibility, commuter behavior, and mechanical reliability, while accounting for broader economic and social trade-offs. This framework serves as the foundation for the synthesis-driven methodological approach introduced in Section 1.2, which combines quantitative modeling evidence, mechanical engineering insights, and contextual analysis to provide a clear path for Kampala's low-carbon transportation future.

1.2. Methodological approach

This study adopts an integrative, synthesis-based methodological approach rather than an experimental or simulation-driven design. The objective is to develop a mechanically grounded conceptual architecture for Kampala's transition to electric Bus Rapid Transit (e-BRT) at a stage when the city has no operational e-bus fleet and empirical datasets remain limited. Accordingly, the analysis draws on established mechanical engineering principles, comparative international benchmarks, and insights from Uganda's transport–energy context.

This framework builds on three modeling pillars: KAMPALA-TIMES, demonstrating up to 40 % emissions reductions; KLAP-TIMES, showing reliability- and cost-driven modal shifts via VTT/TTF dynamics; and GKMA-TIMES–CGE, linking energy optimization to GDP, welfare, and employment. Together, they establish robust system, behavioural, and economic foundations.

This study extends that continuum by introducing a fourth analytical pillar: mechanical systems optimization, the critical operational layer required to translate modelled potential into real-world performance. The analysis identifies context-specific torque thresholds for Kampala's gradients, thermal management requirements for lithium-ion batteries in equatorial heat, and regenerative braking behaviors in congested, stop-and-go traffic. While not producing new empirical data, the study establishes engineering benchmarks that can guide future pilot testing and integration into bottom-up or hybrid modeling frameworks.

By embedding mechanical reliability within broader policy, infrastructure, and grid-planning considerations, Kampala is positioned as a "living laboratory" where engineering performance, commuter responsiveness, energy-system viability, and macroeconomic resilience intersect.

This methodological integration underpins the paper's structure: Section 2 details the mechanical optimization parameters required for reliable electric bus operation. Section 3 connects these parameters to grid integration, corridor design, and governance frameworks. Section 4 synthesizes the multidimensional co-benefits that result when mechanical reliability and policy readiness come together. Section 5 describes research and innovation frontiers, including digital twins and V2G integration. Section 6 presents the study's limitations and implementation risks, and Section 7 offers the conclusions and policy outlook.

2. Mechanical optimization of electric buses

Mechanical optimization underpins credible e-bus deployment in Kampala. Modeling shows emissions cuts of up to 40 % and GDP-intensity reductions above 60 %, but achieving these gains depends on drivetrain torque, thermal battery stability, and regenerative braking—without which reliability, commuter uptake, and macroeconomic benefits cannot materialize.

2.1. Drivetrain design and terrain adaptation

Kampala's radial corridors, such as Jinja Road, Entebbe Road, Hoima Road, Mityana Road, and Masaka Road, are characterized by steep gradients and frequent stop-and-go conditions [54]. Under such conditions, electric bus drivetrains require enhanced torque delivery at low speeds to ensure reliability and energy efficiency [20]. While direct-drive motors perform well on flat, constant-speed operations, modeling studies show that two-speed or multi-speed gearboxes can reduce energy consumption by 7–12 % on urban routes with frequent gradients [28,35].

Representative torque–speed estimates for Kampala's terrain indicate that buses require between 2000 and 2400 Nm of continuous wheel torque to maintain 20–30 km/h on 6–8 % gradients [31,48], which are common along Entebbe Road, Masaka Road, Bombo Road, and parts of Jinja Road. This requirement exceeds the nominal torque of many baseline e-bus platforms, reinforcing the need for torque-multiplying gearbox configurations or high-torque permanent-magnet motors.

Integrating these torque thresholds into bottom-up modeling can be achieved by adjusting drivetrain efficiency curves and auxiliary load penalties within the vehicle-technology database, allowing scenario modeling to capture realistic energy use under Kampala's hilly topography.

For Kampala, tailoring drivetrain specifications to hilly terrains is not only a technical necessity but also a behavioral imperative: if buses stall on inclines or suffer excessive downtime, the commuter confidence gains modeled in KLAP-TIMES scenarios will not materialize. Engineering solutions must therefore be evaluated not in isolation but as enablers of the mode-shift pathways identified in our bottom-up models. (Fig. 1)

2.2. Battery thermal management in hot climates

Battery degradation and thermal instability are significant challenges in Kampala's equatorial climate, where average daytime temperatures range from 27 to 30 °C [14,56]. Empirical studies indicate that passive or air-cooled systems, commonly employed in temperate climates, tend to accelerate capacity fade in hot urban environments [37]. By contrast, liquid cooling systems can extend battery lifespan by 20–30 % [49], while hybrid approaches that integrate phase-change materials or thermo-electric loops offer additional resilience at moderate cost.

Indicative thermal-load analysis shows that maintaining lithium-ion pack temperatures within the safe 20–35 °C window requires 3–5 kW of continuous cooling during daytime operations in Kampala, especially during mid-day peak when thermal soak increases [10,13,32]. This cooling demand translates into a 4–6 % auxiliary energy overhead, which should be reflected in projected route energy consumption and lifecycle cost estimates.

In bottom-up modeling frameworks, these thermal-management penalties can be operationalized by adjusting technology-specific auxiliary load coefficients and degradation rates, ensuring that long-term system scenarios reflect battery wear under hot-climate conditions rather than generic temperate-environment assumptions.

Drawing from the CRT scenario in the metro integration study,

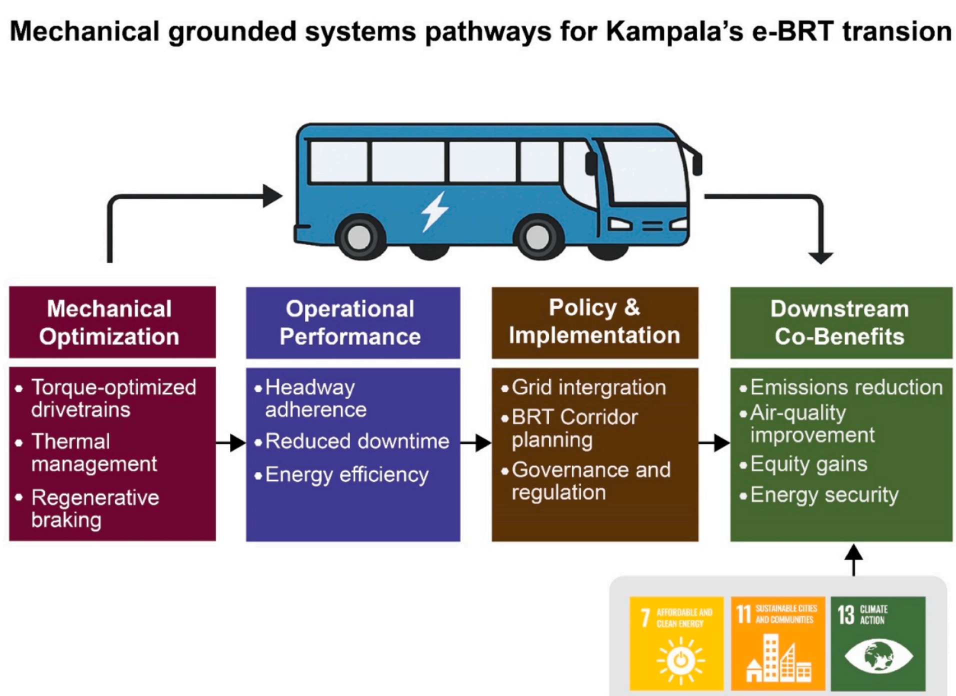

Fig. 1. Graphical Abstract: Mechanically grounded systems pathway for Kampala's e-BRT transition.
The unified graphical abstract illustrates how torque-optimized drivetrains, hot-climate battery thermal management, and regenerative braking establish the mechanical reliability required for dependable e-bus operations. Reliable operations enable effective policy design, BRT corridor integration, and coordinated grid upgrades, which together generate downstream co-benefits including reduced emissions, improved air quality, energy security, equity gains, and progress toward Uganda's NDC and SDG commitments.

where long-term decarbonization depends on durable electrified fleets, it is evident that inadequate thermal management would erode system reliability and fiscal sustainability. Regional R&D partnerships, particularly with universities and local industries, could therefore pioneer cost-effective cooling innovations tailored to Uganda's climatic conditions, positioning Kampala as a continental testbed for optimizing hot-climate batteries.

2.3. *Regenerative braking and stop-and-go efficiency*

Kampala's traffic environment, characterized by chronic congestion, frequent pedestrian interactions, and regular signalized stops, creates favorable conditions for regenerative braking systems. International research indicates that regenerative braking can recover up to 30 % of kinetic energy in dense urban corridors [1]. However, recovery rates depend on algorithm tuning and fleet duty cycles.

A parametric assessment of Kampala's duty cycles indicates realistic regenerative energy recovery levels of 18–26 % for mixed-traffic BRT operations, depending on passenger load, deceleration frequency, and average cruising speeds of 18–25 km/h, consistent with the findings of Szumska & Jurecki [51] and Xu et al. [12]. These recovery ranges can directly decrease per-trip energy consumption by 2–3 kWh/km on specific high-stop corridors such as the Nakawa–CBD and Busega–Nateete links.

In bottom-up modeling, improvements in regenerative braking can be represented by adjusting battery-to-wheel efficiency parameters, allowing scenario techniques to identify the incremental gains in operational efficiency from optimized braking algorithms.

Digital twin applications, already used in our KLAP-TIMES framework to examine commuter flows, can also be expanded to pilot e-bus fleets. By adjusting regenerative braking algorithms to suit Kampala's mixed-traffic BRT conditions, operators can reduce brake wear, improve energy recovery, and increase fleet uptime. Integrating digital optimization with mechanical systems demonstrates the engineering–behavior connection: predictive maintenance and optimized braking cycles directly support commuter reliability, decrease waiting times, and strengthen the mode-shift dynamics captured in our behavioral models.

Drivetrain design, battery cooling, and regenerative braking are foundational to credible e-BRT deployment, linking system-level ambitions to commuter-level reliability. Together with evidence from KAMPALA-TIMES, KLAP-TIMES, and GKMA-TIMES–CGE, these mechanical parameters integrate engineering resilience with behavioural responses and macroeconomic stability, forming a coherent framework for Kampala's transport decarbonization. (Table 1)

2.4. *Mechanics as the linking layer between modeling and implementation*

Kampala's transport–energy models—KAMPALA-TIMES, KLAP-TIMES, and the GKMA-TIMES–CGE hybrid—offer strong insights into system feasibility, commuter behaviour, and macroeconomic impacts, yet they implicitly assume electric-bus reliability. We position mechanical performance as the essential linking layer between modeled decarbonization pathways and practical implementation. Drivetrain torque adequacy, battery thermal stability in hot equatorial conditions, and regenerative-braking efficiency determine whether fleets can operate consistently on steep gradients, under variable loads, and in congested environments. These engineering parameters shape service punctuality and headway adherence, which drive commuter adoption, and underpin fleet availability, maintenance predictability, and long-term battery life, all necessary for macroeconomic gains projected in TIMES–CGE scenarios. Mechanical optimization thus mediates system-level ambition, behavioural responsiveness, and economy-wide benefits. Fig. 2 synthesizes this causal chain, illustrating how torque, thermal management, and regenerative braking influence operational reliability—fleet uptime, energy efficiency, and route performance—and thereby condition the effectiveness of corridor design, regulatory frameworks, and procurement policies. By embedding mechanical variables within a broader systems and policy architecture, the study proposes how engineering reliability enables decarbonization outcomes and translates Kampala's modeled transition pathways into tangible, scalable mobility improvements.

The quantified mechanical requirements outlined in Section 2, torque thresholds for gradient performance, cooling loads necessary for thermal stability, and regenerative-braking recovery ranges translate directly into infrastructure, grid, and governance considerations. These engineering parameters are not isolated technical notes; they determine depot charging architecture, grid reinforcement needs, BRT corridor design, and the operational standards required for fleet procurement. In this way, mechanical reliability becomes the foundational constraint around which policy, infrastructure investment, and institutional coordination must be organized. Section 3, therefore, builds on these engineering insights by examining how grid integration, corridor planning, and governance frameworks must evolve to support mechanically credible e-BRT deployment in Kampala.

Table 1
Comparative insights: Metro, commuter behavior, macroeconomic synergy, and e-BRT optimization in Kampala.

Dimension	Metro integration study (KAMPALA-TIMES)	KLAP-TIMES commuter behavior study	GKMA-TIMES–CGE hybrid synergy	This Perspective: e-BRT mechanical optimization (revised)
Analytical focus	System-level energy and emissions pathways	Mode choice under VTT/TTF constraints	Energy–economy–environment feedbacks	Vehicle-level mechanics (torque, thermal load, regen braking) as operational enablers of system feasibility
Decarbonization potential	Up to 40 % CO_2 reduction under CRT	50 % reduction trajectory by 2060 under KSS	>60 % reduction in CO_2 intensity of GDP under Lutta	25–40 % CO_2 reduction if ~40 % of matatus replaced by e-BRT, conditional on drivetrain torque sufficiency, thermal stability, and regen efficiency
Infrastructure scale	Metro/light-rail corridors as system backbone	Multimodal network with commuter response	Metro-anchored electrification + supply-mix optimization	Incremental BRT corridors along radial roads (Jinja, Bombo, Masaka, Entebbe)
Key operational enabler	Grid expansion + renewable portfolio	Travel-time savings and reliability	Low-carbon electricity + structural economic reallocation	Torque-optimized drivetrains (2000–2400 Nm), battery thermal management (3–5 kW cooling load), adaptive regenerative braking (18–26 % recovery)
Modeling capability	Robust scenario modeling but no vehicle-level mechanical systems detail	Captures behavioral elasticity but not mechanical engineering constraints	Captures macroeconomic trade-offs but not mechanical variables	Mechanical parameters can be operationalized in bottom-up modeling via efficiency curves, auxiliary load coefficients, degradation factors, and gradient-penalty adjustments
Policy implication	High capital investment; long-term buildout	Policy must enhance reliability to shift demand	Aligns abatement with GDP, welfare, and industrial pathways	Mechanical reliability becomes a prerequisite for policy credibility and commuter adoption
Knowledge gap addressed	System feasibility; limited on bus operations	Commuter dynamics; no link to mechanical constraints	Economy-wide impacts; no operational detail	Bridging system visions, commuter needs, and economy-wide goals through engineering-specific operational realism

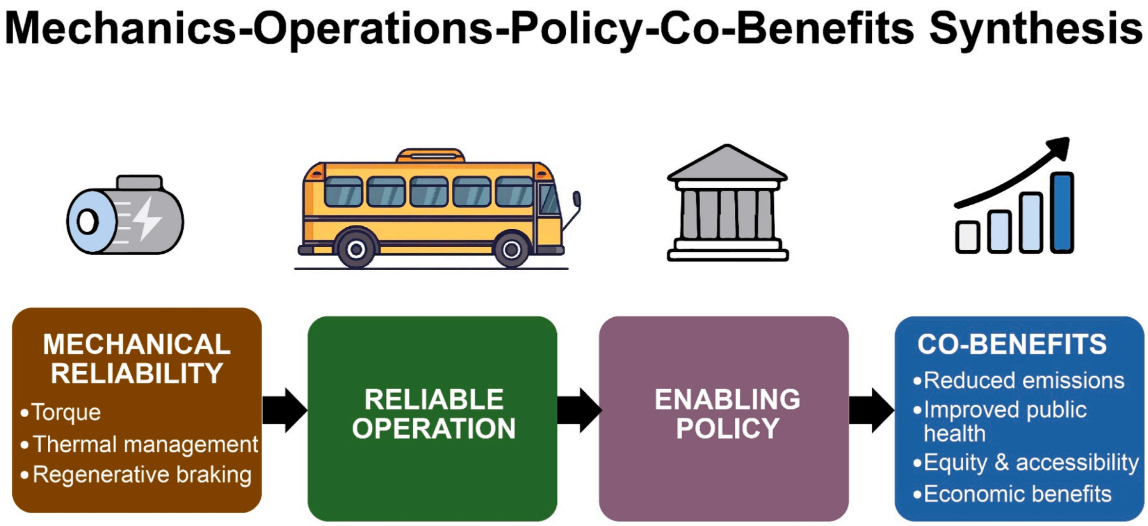

Fig. 2. Conceptual systems pathway- mechanical optimization-operational reliability- multidimensional co-benefits.

3. Policy and infrastructure integration for e-BRT

3.1. Grid integration and charging infrastructure

The electrification of Kampala's bus fleet requires careful planning for charging infrastructure and integration with Uganda's electricity grid. Uganda's grid is currently mainly powered by hydropower, with an installed capacity of about 1250 MW and a peak demand of just under 800 MW [26]. While this offers a low-carbon baseline, local distribution bottlenecks, reliability issues, and seasonal fluctuations pose challenges for electric bus operations. The KAMPALA-TIMES analysis highlighted that electrified transportation is achievable under scenarios like the Carbon Reduction Target (CRT), but only if supply-side stability is ensured [25].

Charging models usually fall into two categories: depot-based charging and on-route opportunity charging. Depot-based charging, overnight charging at bus depots, is easier to manage but requires higher battery capacities, which adds weight and increases costs. In contrast, on-route charging reduces battery size requirements but demands a significant upfront investment in fast-charging infrastructure at terminals or key bus stops.

In Kampala, a hybrid model is the most practical option. BRT depots located along key corridors (e.g., Busega, Nakawa, Kajjansi, Matugga, and Namboole) could be equipped with solar PV and battery storage systems to supplement grid supply and ensure resilience during outages. Modeling studies show that solar-assisted depots can reduce peak grid demand by 20–30 %, while also providing cost savings on electricity tariffs [30]. For Uganda, this integration aligns with the National Renewable Energy Policy [22], which prioritizes distributed solar generation for urban resilience. Such hybrid solutions provide a technical basis for mobilizing climate finance.

3.2. e-BRT corridor planning and spatial optimization

Kampala's urban layout is mainly radial, with corridors like Jinja Road, Entebbe Road, Bombo Road, and Masaka Road guiding commuters into the Central Business District. Although the Kampala Capital City Authority (KCCA) has proposed BRT systems for over 10 years, progress has been slow due to funding constraints and fragmented governance. Based on the KAMPALA-TIMES findings, which highlighted multimodal integration anchored by high-capacity electrified transit, e-BRT corridors offer a practical, cost-effective complement to long-term metro projects.

International experience bolsters this argument. In Dar es Salaam, dedicated BRT lanes cut travel time by 40 % and fuel use by 35 % [19]. Modeling for Kampala indicates that replacing 40 % of matatu operations with e-BRT could reduce CO_2 emissions by 25–40 % by 2040, depending on fleet size and charging methods [24]. The behavioral evidence from KLAP-TIMES shows that such gains in travel time and reliability are crucial for commuters: under the Kampala Sustainable Scenario (KSS), VTT and TTF constraints led to a significant shift toward electrified mass transit [24]. Therefore, corridor planning is not just about civil works—it enables the behavioral flexibility captured in bottom-up models.

From a mechanical engineering perspective, dedicated BRT lanes further optimize driving cycles, reducing drivetrain stress and improving regenerative braking recovery. Infrastructure and mechanics are therefore interconnected: infrastructure sets the operational conditions where mechanical optimizations can achieve their maximum energy and reliability benefits.

3.3. Governance, regulation, and financing models

A key lesson from our previous studies is that technical feasibility and commuter acceptance are not sufficient without strong governance frameworks. Uganda's transport sector remains highly fragmented, with overlapping responsibilities among KCCA, the Ministry of Works and Transport (MoWT), and deeply rooted informal transport operators. The KLAP-TIMES results showed that commuters are willing to switch to more sustainable modes when reliability is assured, but this shift cannot happen in a policy vacuum.

International examples provide valuable governance models. Bogotá's TransMilenio delegated bus operations to private consortia while keeping public control over fare collection and infrastructure [45]. Shenzhen achieved full electrification of its bus fleet by combining strong mandates with government subsidies [57], while Nairobi's e-mobility pilots use donor-supported PPP frameworks to lower investor risk [44]. For Kampala, a PPP structure appears most promising: government investment in infrastructure (lanes, depots, charging hubs) paired with private fleet operation. The draft National E-Mobility Strategy (2024) already suggests innovative financing options, including carbon credits and green bonds [22]. Incorporating e-BRT projects into the Green Climate Fund or Article 6 carbon markets could significantly reduce capital risk while supporting Uganda's NDC commitments.

3.4. Modeling policy scenarios

Policy scenario modeling links engineering feasibility, commuter behaviour, and economy-wide outcomes. Building on KAMPALA-TIMES (40 % emissions reduction) and KLAP-TIMES (reliability-driven modal shifts), this study positions mechanical optimization as the operational layer enabling credible e-BRT transitions. Incorporating drivetrain efficiency, thermal stability, and regenerative braking gains into bottom-up modeling enables the analysis of three pathways: continued diesel growth, partial e-BRT adoption (~20 % by 2040), and full deployment (50–60 % by 2040). Coupled with a recursive dynamic CGE analysis, this integrated framework supports technically realistic and economically sustainable mobility policy.

Kampala's policy and infrastructure modeling shows that system feasibility, commuter responsiveness, and macroeconomic gains depend on reliable mechanical performance. Mechanical optimization and infrastructure readiness form the missing link translating modeling insights into operational credibility, enabling BRT corridors, resilient grids, and inclusive governance to drive a sustainable, equitable, low-carbon transport transition.

The policy and infrastructure configurations discussed in Section 3 only achieve their intended impact when paired with the mechanical reliability established in Section 2. Electrified BRT corridors yield meaningful environmental, economic, and equity gains only when buses maintain torque performance on steep inclines, avoid thermal derating in hot conditions, and maximize regenerative energy recovery in congestion. These operational factors directly affect service frequency, commuter satisfaction, fleet availability, and long-term fiscal sustainability. Section 4, therefore, synthesizes these linkages by showing how mechanically credible and policy-enabled e-BRT systems generate multidimensional co-benefits—from public health improvements and economic resilience to social equity and alignment with Uganda's climate commitments.

4. Co-Benefits of e-BRT transition

To clarify how the co-benefits discussed in this section arise not as automatic consequences of electrification but as downstream outcomes mediated through engineering and operational performance, Fig. 3 synthesizes the causal pathway linking mechanical optimization to improved service reliability and, ultimately, to environmental and social benefits. By visually separating direct mechanical effects from operational enhancements and broader co-benefits, the study reinforces the central argument that Kampala's e-BRT system can achieve meaningful public-health, equity, and resilience gains only if the underlying fleet mechanics consistently support dependable, high-quality service.

4.1. Air quality and public health

One of the most immediate and visible benefits of switching to electric BRT is improving urban air quality [29]. Kampala's $PM_{2.5}$ levels often surpass WHO guidelines by five times, with diesel minibuses and motorcycle taxis identified as the main contributors [40]. Replacing even a small portion of these vehicles with electric buses would significantly cut black carbon and NO_x levels. Lessons from our KAMPALA-TIMES metro study show that fully electrifying public transport corridors significantly reduces urban emissions [25]. Meanwhile, KLAP-TIMES findings confirm that commuters respond well to service quality improvements when health benefits and reliability are clear [24].

Extrapolating from Shenzhen, where full fleet electrification is projected to reduce transport-related $PM_{2.5}$ emissions by 50 % within a decade [8], Kampala could anticipate significant public health gains. Potential savings of $50–70 million annually could accrue from avoided healthcare costs associated with asthma, cardiovascular disease, and premature mortality. This represents both a public health and an economic dividend. Crucially, the mechanical optimizations outlined earlier, robust battery cooling, torque-sensitive drivetrains, and regenerative braking ensure fleet reliability, preventing a return to polluting paratransit when electric buses fail. In this sense, public health benefits are structurally tied to engineering reliability.

4.2. Economic resilience and energy security

Transport electrification also provides co-benefits such as enhanced economic resilience [15]. Uganda spends over $1.5 billion annually on petroleum imports, which continuously drain its foreign exchange reserves [36]. Switching to electric BRT reduces reliance on imported oil, shielding the economy from fluctuating global fuel markets. Our KAMPALA-TIMES scenarios confirmed that decarbonized transportation pathways not only lower emissions but also reduce exposure to fossil-fuel shocks [25].

The integration of solar-PV-assisted depots with storage enhances

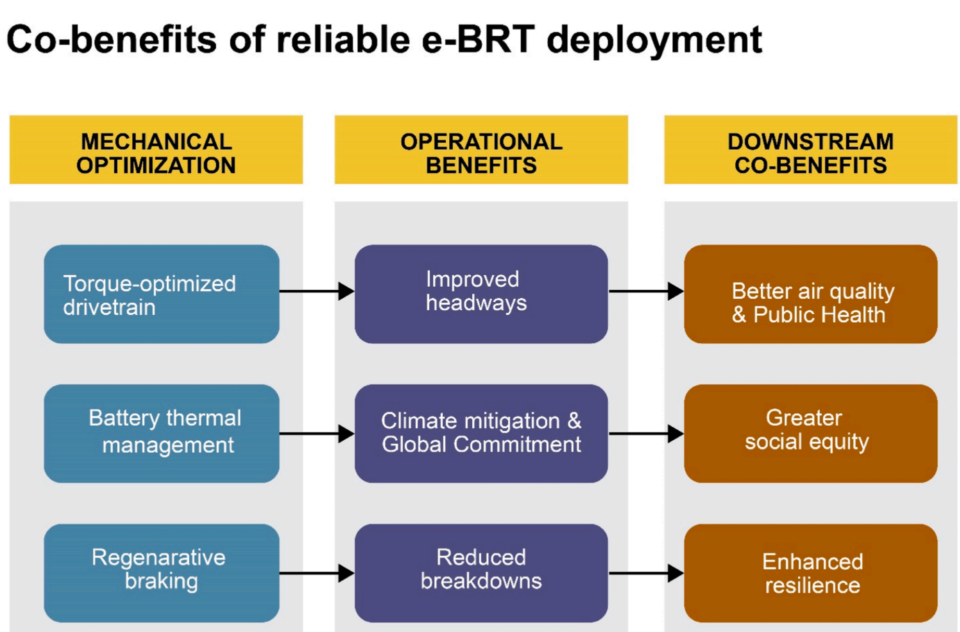

Fig. 3. Mechanically mediated pathway to downstream co-benefits in Kampala's e-BRT system.

energy security, a key focus in Kimuli et al. [27]. Pilot studies across Africa show that the operating costs per kilometer for e-buses are 30–40 % lower than those of diesel buses [39], creating room for reinvesting in fleet expansion and infrastructure improvements. Also, as KLAP-TIMES demonstrated, consumer acceptance depends on affordability. Lower operating costs can be converted into stable, affordable fares, boosting commuter confidence.

Furthermore, local assembly and component manufacturing, as demonstrated in Kenya and Rwanda, could help develop new industrial value chains in Uganda [6,42]. Opportunities include battery recycling, power electronics, and predictive maintenance technologies—supporting Uganda's National Industrial Policy (2020) and opening the door to job creation and skills development.

4.3. Social equity and mobility justice

Equity remains a key issue in sustainable transport transitions. Kampala's current paratransit system is unreliable, unsafe, and costly, especially relative to household incomes: women and low-income commuters in particular face long waits, harassment, and safety concerns. The KLAP-TIMES analysis emphasized that commuters value time savings and reliability when choosing modes [24]. If e-BRT can provide predictable schedules, secure boarding, and fixed fares, it directly addresses these behavioral concerns and promotes mobility justice.

Case studies from Bogotá [43] and Mexico City [5] confirm that BRT systems improve access to jobs and education for marginalized groups. For Kampala, spatial modeling shows that BRT deployment could cut average commute times by 30–40 %, offering significant benefits for low-income households [24]. Additionally, electrification reduces noise pollution [4,17], a rarely measured but socially important factor in making neighborhoods more livable. Therefore, mechanical optimization, infrastructure planning, and commuter equity come together as a unified path to inclusive mobility.

4.4. Climate mitigation and global commitments

At the macro level, electric BRT directly supports Uganda's Nationally Determined Contribution, which pledges to cut GHG emissions by 24.7 % by 2030 compared to business-as-usual [23]. Transport is a key sector, but the pathways for implementation remain unclear. As shown in the CRT scenario of the KAMPALA-TIMES study, electrified mass rapid transit (MRT) corridors are crucial for achieving significant reductions [25]. Well-optimized e-BRT fleets therefore offer a clear, measurable strategy for meeting international goals under the Paris Agreement.

Furthermore, Kampala's transition could establish the city as a regional living laboratory, promoting technology transfer and South–South collaboration. Joining the Article 6 carbon markets could unlock climate finance streams, supporting long-term investment in fleet procurement and infrastructure development.

4.5. Synergies with broader sustainable development goals (SDGs)

The co-benefits of electric bus rapid transit (e-BRT) deployment in Kampala go beyond transportation and climate policies, linking the transition to the broader Sustainable Development Goals (SDGs) framework. At the most immediate level, reducing tailpipe emissions and related air pollutants directly supports SDG 3 (Good Health and Well-being) by reducing the incidence of respiratory and cardiovascular diseases among urban populations exposed to high levels of $PM_{2.5}$ and NO_x.

From an energy systems perspective, integrating solar-assisted depots and distributed storage solutions into the charging infrastructure advances SDG 7 (Affordable and Clean Energy) by linking e-mobility with renewable energy and reducing grid peak load. These connections not only reduce operational costs but also enhance energy resilience in a city often affected by localized grid instability.

The transition also has important effects on employment and industrial growth. By creating new opportunities in local bus assembly, battery recycling, and predictive maintenance, e-BRT supports SDG 8 (Decent Work and Economic Growth) and promotes SDG 9 (Industry, Innovation, and Infrastructure) through modernizing Uganda's urban mobility systems and encouraging technology transfer. These industrial paths align with Uganda's broader national industrial policy and offer opportunities to enhance regional competitiveness.

Equally important are the social equity dimensions. By providing reliable, affordable, and safe mobility, electric BRT corridors directly support SDG 11 (Sustainable Cities and Communities), thereby improving access to employment and education opportunities for low-income groups, women, and youth. Finally, at the global level, Kampala's transportation decarbonization directly advances SDG 13 (Climate Action) by reducing greenhouse gas emissions in line with Uganda's Nationally Determined Contributions (NDCs) and strengthening the country's credibility in international climate negotiations.

By integrating e-BRT into this multidimensional SDG framework, Kampala's low-carbon transportation strategy shifts from simply being an engineering or policy reform project to becoming a comprehensive development initiative. This broader perspective can help attract not only climate finance but also diverse donor and private-sector investments, thus anchoring the city's mobility transformation in both local development goals and global sustainability commitments.

5. Innovation and research frontiers

5.1. Kampala as a living laboratory for low-carbon mobility

Kampala demonstrates both the urgency and the potential of transportation decarbonization in the Global South. The city's rapid motorization, reliance on informal paratransit, and high climate vulnerability create conditions where innovative solutions can deliver significant benefits. Unlike cities in the Global North, which are limited by legacy infrastructure, Kampala is at a crucial point: the lack of entrenched mass rapid transit systems offers an opportunity to directly adopt electrified BRT corridors, smart charging depots, and renewable grid integration.

KAMPALA-TIMES shows that electrified high-capacity transit can cut emissions by 40 %, while KLAP-TIMES demonstrates that reliability-driven commuter shifts enable a 50 % reduction by 2060. The GKMA-TIMES–CGE hybrid extends these insights, revealing that deep decarbonization reduces GDP carbon intensity by over 60 % and strengthens welfare and sectoral resilience(Fig. 4).

The current study adds a fourth layer: vehicle-level mechanics—such as drivetrain optimization, thermal management, and regenerative braking—that translate system, behavior, and economic benefits into tangible real-world outcomes. Together, these elements turn Kampala into a living laboratory where system, behavioral, macroeconomic, and mechanical innovations combine to foster sustainable and equitable urban mobility.

5.2. Coupling times with CGE macroeconomic modeling

The TIMES framework captures techno-economic feasibility but overlooks macroeconomic feedbacks affecting welfare, employment, and fiscal stability. KAMPALA-TIMES showed that electrified metro corridors can cut emissions by 40 %, while KLAP-TIMES demonstrated that affordability and reliability drive commuter shifts. However, feasibility and behaviour alone are insufficient for policy credibility. The GKMA TIMES/CGE hybrid addressed this gap, revealing that deep decarbonization improves GDP and household welfare, reduces GDP carbon intensity by 60.6 % by 2050, and exposes distributional effects requiring targeted compensatory policies.

For Kampala's e-BRT transition, this coupling allows a set of policy-relevant questions to be examined with analytical rigor. A key issue

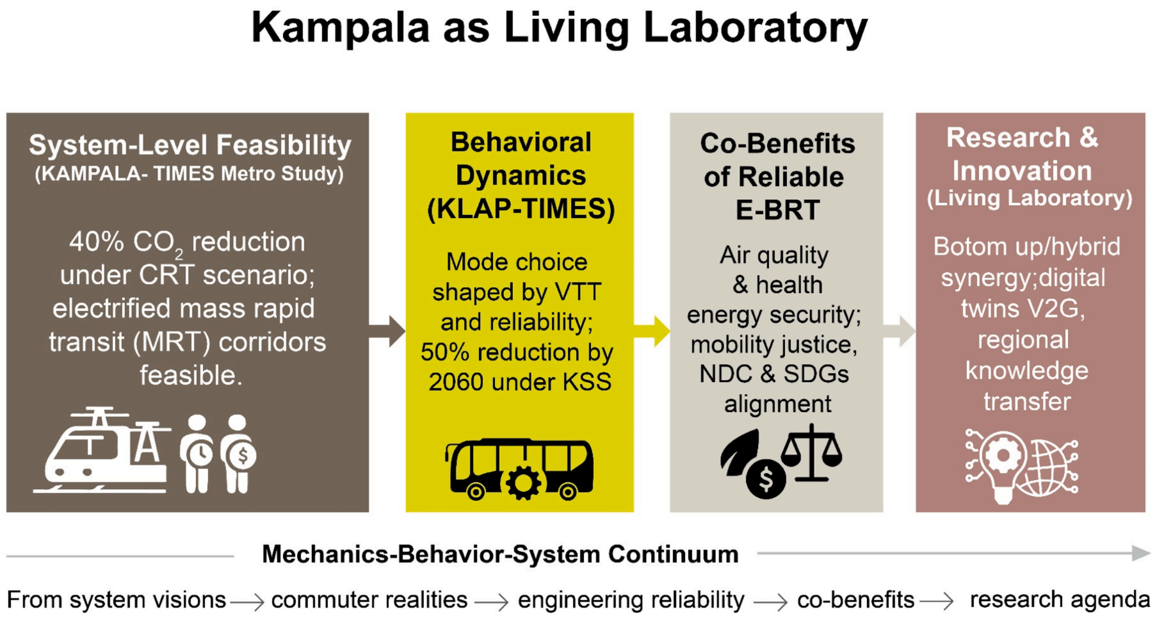

Fig. 4. Conceptual framework.

involves the GDP and trade balance effects of replacing imported diesel with domestically produced e-BRT, which could lessen foreign exchange pressures while boosting local energy markets. Equally important is the impact of fare structures on household welfare across income groups, especially for low-income commuters identified in KLAP-TIMES as most sensitive to travel costs. Lastly, the transition raises labor-market considerations, requiring strategies to mitigate employment losses in the paratransit sector through retraining programs that prepare workers for roles such as e-bus driving, fleet maintenance, and battery recycling. Collectively, these questions highlight how an integrated TIMES–CGE framework links engineering optimization with broader economic and social outcomes, ensuring Kampala's mobility transition is both technically feasible and socially fair.

By integrating mechanical optimization parameters (e.g., efficiency improvements from regenerative braking, extended battery life from thermal management) into TIMES and then feeding these outputs into CGE, policymakers obtain a comprehensive view: not just emissions reductions but also economic growth, employment, and equity. This combined TIMES–CGE hybrid approach ensures that Kampala's e-mobility strategy is grounded in engineering realism, behavioral acceptance, and macroeconomic sustainability, complementing rather than replacing the insights from the GKMA-TIMES/CGE hybrid study, KAMPALA-TIMES, and KLAP-TIMES.

5.3. Digital twins and predictive maintenance

The deployment of digital twins, dynamic, real-time virtual replicas of buses and fleets, represents a frontier in predictive optimization [11]. In Kampala, digital twins could be used to optimize regenerative braking in stop-and-go traffic, thereby maximizing energy recovery and improving overall efficiency [16,50]. They can also help predict battery degradation under equatorial heat, reducing unplanned downtime and extending battery lifespans. Additionally, digital twins enable more effective preventive maintenance scheduling, thereby increasing fleet availability and enhancing commuter trust in the reliability of electric BRT systems [3,33].

By aligning mechanical reliability with commuter expectations, as highlighted in the KLAP-TIMES analysis, digital twins create a strong feedback loop in which engineering precision directly drives behavioral shifts toward low-carbon transportation options. Beyond their operational uses, datasets produced by digital twin technologies could position Kampala as a regional leader in transportation AI innovation, especially when combined with higher education institutions' established strengths in systems modeling and energy optimization.

5.4. Vehicle-to-Grid (V2G) integration

Another promising research frontier is vehicle-to-grid (V2G) technology, where electric buses act as mobile energy storage units that can feed electricity back into the grid during peak demand. Given Uganda's high reliance on hydropower, seasonal fluctuations in water inflows can strain the grid. Strategically located e-bus depots with V2G functions could provide peak-shaving services, thus improving energy security.

Pilot studies in Europe [7] and Asia [46] show that aggregated bus fleets can deliver hundreds of megawatt-hours of flexible capacity. In Kampala, V2G could be tested at depots near industrial parks or universities, where daytime charging aligns with solar PV generation. This synergy between mechanical systems (bus batteries) and energy systems (grid flexibility) highlights the cross-sector innovation potential of e-BRT fleets. As demonstrated in the KAMPALA-TIMES scenarios [25], transportation electrification interacts dynamically with Uganda's power mix—V2G adds resilience to this equation.

5.5. Research capacity and collaborative innovation

Realizing Kampala's low-carbon transport shift requires not only technological innovation but also stronger institutional capacity and collaborative research ecosystems. Higher Institutions of learning, with their expertise in mechanical engineering, energy systems modeling, and urban planning, are well-positioned to lead a multidisciplinary Urban Transport Innovation Lab. This platform could combine technical experimentation, policy analysis, and behavioral research, while partnering internationally with organizations such as IEA-ETSAP, the World Resources Institute, and regional centers of excellence to offer advanced modeling tools, capacity-building, and financial support. Incorporating research modules into pilot e-BRT corridors covering drivetrain performance, battery thermal reliability, and commuter satisfaction would promote ongoing learning, knowledge sharing, and local ownership, helping to bridge the gap between research and implementation.

KAMPALA-TIMES established the system-level feasibility of cutting emissions by 40 %; KLAP-TIMES showed that commuter shifts depend on affordability and reliability; and the GKMA-TIMES–CGE hybrid revealed over 60 % reductions in GDP intensity with welfare gains. This study extends that continuum by identifying mechanical optimization as the operational enabler of credible, reliable e-BRT outcomes.

Fig. 4 presents the Mechanics–Behavior–System Continuum, showing how system-level feasibility (40 % emissions reduction in KAMPALA-TIMES) and commuter responsiveness (50 % reduction by 2060 in KLAP-TIMES) depend on drivetrain torque, thermal battery cooling, and regenerative braking, positioning Kampala as a living laboratory for Africa's low-carbon mobility transitions.

6. Limitations and implementation risks

This study provides a mechanically grounded conceptual framework for Kampala's e-BRT transition, but several limitations temper its applicability. Because the city lacks an operational e-bus fleet, torque, thermal load, and regenerative braking estimates are derived from engineering benchmarks rather than local duty-cycle data; pilot deployments, digital twins, and instrumented testing will be required for empirical validation. Moreover, although the study outlines how mechanical parameters can be incorporated into future transport–energy models, no empirical bottom-up/hybrid modeling was performed, leaving uncertainties in drivetrain derating, battery degradation, and auxiliary thermal loads unquantified. Implementation challenges also stem from fragmented governance across KCCA, MoWT, UMEME, ERA, MEMD, and private operators, as well as grid-capacity constraints in distribution networks and urban substations. Macroeconomic volatility—in tariffs, exchange rates, and financing conditions—further threatens lifecycle affordability and renewal planning. Behavioural sensitivities identified in KLAP-TIMES suggest that unreliable operations could undermine expected modal shifts, weakening emissions and equity benefits. As a conceptual study, the framework provides directional insight rather than quantified projections. Advancing Kampala's e-BRT ambitions will require phased piloting, cross-agency coordination, and iterative data collection, supported by digital-twin analytics, thermal and torque profiling, and degradation-aware modeling to ensure that system-level ambitions translate into reliable, equitable, and resilient mobility outcomes.

7. Conclusion and policy outlook

Kampala's shift to electric buses and BRT links engineering reliability, commuter behaviour, and energy management. KAMPALA-TIMES shows up to 40 % emissions reduction, KLAP-TIMES highlights reliability-driven modal shifts toward a 50 % cut by 2060, and GKMA-TIMES–CGE demonstrates GDP gains, welfare protection, and reduced oil dependence. This study advances the discourse by identifying mechanical optimization as the missing bridge between modelled feasibility and real-world implementation. Drivetrain capability on steep corridors, robust thermal management in equatorial heat, and regenerative braking in congested traffic are essential for maintaining reliability, headway adherence, and lifecycle performance—conditions without which neither the projected emissions reductions nor behavioural shifts can be realized. Mechanically credible e-BRT operations also generate broader co-benefits: cleaner air, reduced public health burdens, strengthened energy security, improved equity, and enhanced alignment with Uganda's NDCs and SDGs 3, 7, 8, 9, 11, and 13. Thus, e-BRT is not merely a transport intervention but a resilience and development strategy requiring integrated engineering design, coordinated governance, and targeted fiscal planning.

7.1. Policy recommendations

To operationalize this vision, we recommend:

1. **Establish mechanically informed procurement and operational standards**: Uganda should mandate drivetrain torque benchmarks for hilly corridors, thermal-management specifications suited to equatorial temperatures, and regenerative-braking performance thresholds in all e-BRT fleet tenders. UNBS, MoWT, and KCCA should codify these standards to ensure that mechanical reliability underpins service quality and modal-shift expectations.
2. **Sequence e-BRT corridor deployment through data-driven, route-specific planning**: KCCA should implement a phased, corridor-by-corridor rollout strategy based on traffic density, gradient profiles, feeder network integration, and grid accessibility. This approach ensures that high-priority corridors (Jinja Road, Bombo Road, Entebbe Road, Masaka Road) receive early investments tailored to their mechanical and operational needs.
3. **Implement integrated charging infrastructure planning with utility coordination**: UMEME, UETCL, and KCCA should jointly develop depot and on-route charging strategies that combine grid reinforcement, solar-PV hybridization, and substation upgrades. Planning should account for verified cooling loads, auxiliary energy demand, and fleet availability requirements identified in Section 2.
4. **Establish a multi-agency e-BRT governance framework**: A dedicated Kampala e-Mobility Implementation Unit, authorized by KCCA, MoWT, MEMD, ERA, and utilities, should coordinate fleet procurement, charging infrastructure design, traffic management, and performance-based PPP contracts to reduce institutional fragmentation and delays in implementation.
5. **Incorporate equity, affordability, and commuter reliability into policy planning**: Fare structures, service frequencies, and feeder networks should explicitly consider low-income commuters, gender-sensitive travel patterns, and vulnerable groups. Reliability metrics related to mechanical performance (headway adherence, fleet uptime, energy efficiency) must be integrated into service-level agreements.
6. **Require all pilots to collect empirical data for TIMES–CGE hybrid synergy and mechanical model updates**: Pilot e-BRT deployments should include digital-twin monitoring, gradient-specific torque profiling, thermal-cycle analysis, and regenerative-braking recovery measurements. These datasets should feed directly into updated hybrid modeling scenarios, allowing for evidence-based scaling and policy refinement.

If pursued decisively, Kampala's e-BRT transition could set a continental example for sustainable mobility, providing lessons for quickly growing cities like Nairobi, Kigali, and Dar es Salaam. By combining engineering excellence with policy innovation and social inclusion, Kampala establishes itself as a living laboratory for the Global South, showing how mechanics, behavior, and systems modeling can work together to achieve local resilience and global climate goals.

CRediT authorship contribution statement

Ismail Kimuli: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **John Baptist Kirabira:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

I am writing to formally declare that the authors of our manuscript titled "Low-Carbon Urban Transportation: A Sustainable Mechanical Systems Optimization for Electric Bus and BRT Fleet Deployment in Kampala" **have** no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The research presented forms part of our broader scholarly inquiry into sustainable energy and transport systems in Uganda, building upon our earlier publications — KAMPALA-TIMES, KLAP-TIMES, and GKMA-TIMES–CGE — which collectively advance the discourse on low-carbon urban mobility transitions. All authors have read and approved the final manuscript and consent to its submission to Transportation Engineering.

Data availability

Data will be made available on request.

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